Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/28/2005
PI Name:	Shaw, Benjamin Ph.D.		
Project Title:	Combustion Experiments in Reduced Gravity with Two-Component Miscible Droplets		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bdshaw@ucdavis.edu	**Fax:**	FY 530-752-4158
PI Organization Type:	UNIVERSITY	**Phone:**	530-752-4130
Organization Name:	University of California, Davis		
PI Address 1:	Department of Mechanical and Aeronautical Engineering		
PI Address 2:			
PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/30/2004	**End Date:**	12/21/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA65A		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The program is both experimental and theoretical in nature. The experiments will be conducted in ground-based facilities and on space-based platforms. Normal-gravity comparison experiments will also be performed. Bi-component droplets will be dispensed, deployed and ignited in reduced gravity. The droplet regression history will be measured by backlighting the droplet. An orthogonal view of the flame provides the flame history. Droplet interiors will be imaged, and flow visualization will be used to provide information on flowfields within droplets. The ground-based experiments will be limited to smaller droplets. The space-based experiments will use the Multi-User Droplet Combustion Apparatus hardware to examine the combustion history of larger droplets with correspondingly larger time and length scales.		

Task Description:	Theoretical and numerical studies will be used to model the combustion of the droplets in reduced gravity. The liquid phase transport processes will be of particular interest. The influence of surface tension driven flows on liquid phase transport will also be studied. The objective of this research is to study the combustion of bi-component miscible droplets in reduced-gravity environments so that buoyancy effects are rendered negligible. Of particular interest are the burning rate (droplet history), the time to flame contraction, potential microexplosions, Marangoni flows, and sooting behaviors. The experiments use liquid mixtures selected to display influences of large component volatility differences and liquid species diffusion on combustion phenomena. The research extends previous research on combustion of pure liquid fuels to mixtures of pure fuels, which more closely approximate practical fuels. Thus, this program should lead to a better understanding of the combustion of practical liquid fuels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research extends previous research on combustion of pure liquid fuels to mixtures of pure fuels, which more closely approximate practical fuels. Thus, this program should lead to a better understanding of the combustion of practical liquid fuels. Combustion phenomena of particular interest are burning rates (droplet histories), transport of mass and energy within droplets, flame size histories, bubbling within droplets, Marangoni flows, and sooting behaviors. The results from this research program are expected to be relevant to design of practical combustion systems that employ liquid fuels such as engines and oil-fired burners, e.g., by enabling the design of devices with higher efficiencies and reduced pollutant emissions. The results should also be useful for improving understanding of combustion phenomena under reduced gravity conditions, which is relevant to spacecraft fire safety.
Task Progress:	Research was performed on to investigate aspects of combustion of isolated droplets burning under reduced gravity conditions. Analysis of reduced-gravity experiments involved digitizing and analyzing video images from experiments on combustion of propanol droplets, initially about 1 mm in diameter, in air at pressures from 0.1 MPa to 1.0 MPa. The results indicate that as the pressure increases, burning rates increase. Droplets did not soot at 0.1 MPa, but strong sooting was observed at 0.3 MPa and above. Theory indicates that increases in burning rates are caused by changes in properties, e.g., the liquid enthalpy of vaporization, as the pressure increases. Also, flames became less spherical at elevated pressures (from buoyancy influences), and data were obtained on flame aspect ratios and sizes as functions of time. Computational modeling involved extending a previous numerical model to predict activity coefficients for propanol-water and propanol-water-glycerol mixtures. Calculations indicate that water absorption is very small (mass fractions generally less than 1 %) for propanol droplets burning in air at pressures ranging from 0.1 MPa to 1.0 MPa. A multidimensional computational model was also developed further. The code is based on overset grid technology and uses several meshes to resolve different parts of a droplet on a fiber. The code was used for mass-transfer studies (without combustion but with surface-tension influences). Surface tension forces were predicted to be important in terms of mass transfer and flowfield characteristics. Analytical modeling focused on predicting influences of fuel pyrolysis on droplet combustion characteristics. Accounting for fuel pyrolysis changes classical expressions for the transfer number because the flame must supply energy for fuel pyrolysis, which requires the flame to be located closer to the droplet. Transfer numbers, burning rates and quasisteady flame standoff ratios can be appreciably reduced when fuel pyrolysis effects are included in the analyses. It was also found that fuel pyrolysis can influence temperature gradients between droplets and flames, which influences formation of soot shells.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Dee V, Shaw BD. "Combustion of propanol–glycerol mixture droplets in reduced gravity." International Journal of Heat and Mass Transfer 2004 Oct;47(22):4857-67. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2004.05.025 , Oct-2004
Articles in Peer-reviewed Journals	Dwyer HA, Shringi D, Shaw BD. "A simulation of a fiber supported droplet." Computational Fluid Dynamics Journal. 2004 Oct;13(3):489-95. , Oct-2004
Presentation	Dakka, S. M.; Shaw, B. D. "Combustion of Propanol Droplets in Reduced Gravity" Presented as a poster at the 30th International Symposium on Combustion, Chicago, IL Jul-2004
Presentation	Shaw, B. D.; Dakka, S. M. "Theory of Influence of Fuel Pyrolysis on Soot-Shell Standoff Ratios in Reduced-Gravity Droplet Combustion" Presented as a poster at the 30th International Symposium on Combustion, Chicago, IL Jul-2004
Presentation	Dwyer, H. A.; Shringi, D.; Shaw, B. D. "A Simulation of a Fiber Supported Droplet" Poster presented at the 30th International Symposium on Combustion, Chicago, IL Jul-2004